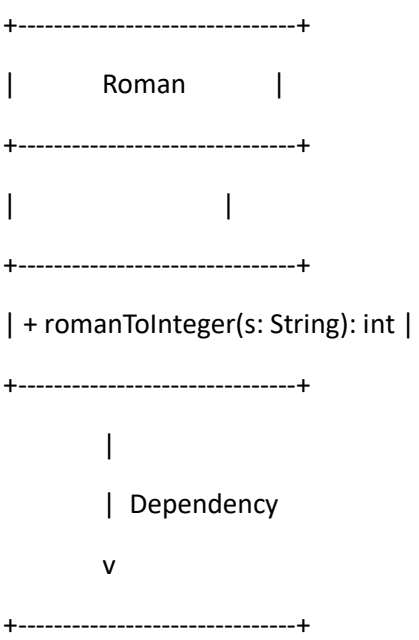
Question:

Define a package number and in that define Roman class and implement romanToInteger() and import the method in another class.

Algorithm:

1. Start.
2. Create a class Roman inside the number package.
3. Define a method romanToInteger(String s) and initialize result = 0.
4. Read the Roman numeral from the user in the Main class.
5. Traverse each character of the Roman numeral from left to right.
6. Find the integer value of the current Roman symbol using a switch statement.
7. If a next character exists, find its integer value.
8. Compare the current value with the next value.
9. If the current value is smaller than the next value, subtract the current value from result.
10. Otherwise, add the current value to result.
11. Repeat until all characters are processed.
12. Return result from the romanToInteger() method.
13. Display the integer value.
14. Stop.

UML Class Diagram:



```
|            Main            |
+-----------------------------+
| - sc: Scanner               |
+-----------------------------+
| + main(args: String[]): void  |
+-----------------------------+
```

Code:

```
package number;


public class Roman {


   // Method to convert Roman numerals to integer
   public static int romanToInteger(String s) {
      int result = 0;


      for (int i = 0; i < s.length(); i++) {
         int value = 0;


         // Get the value of the current Roman symbol
         switch (s.charAt(i)) {
            case 'I': value = 1; break;
            case 'V': value = 5; break;
            case 'X': value = 10; break;
            case 'L': value = 50; break;
            case 'C': value = 100; break;
            case 'D': value = 500; break;
            case 'M': value = 1000; break;
         }


         // Subtract if the current value is smaller
```

```
            // than the next Roman symbol
            if (i + 1 < s.length()) {
                int next = 0;

                switch (s.charAt(i + 1)) {
                    case 'I': next = 1; break;
                    case 'V': next = 5; break;
                    case 'X': next = 10; break;
                    case 'L': next = 50; break;
                    case 'C': next = 100; break;
                    case 'D': next = 500; break;
                    case 'M': next = 1000; break;
                }

                if (value < next) {
                    result -= value;
                } else {
                    result += value;
                }
            } else {
                result += value;
            }
        }

        return result;
    }
}
```

```
import java.util.Scanner;
```

```
import number.Roman;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read the Roman numeral
        System.out.print("Enter Roman numeral: ");
        String s = sc.next();

        // Call the imported method
        int result = Roman.romanToInteger(s);

        // Display the integer value
        System.out.println("Integer value: " + result);

        sc.close();
    }
}
```

Output:

```
Enter Roman numeral: LVIII
Integer value: 58
```